

TRƯỜNG ĐẠI HỌC CÔNG NGHIỆP TP.HCM

Khoa Công Nghệ Điện Tử - Bộ môn Điện Tử Máy Tính

BÀI GIẢNG

**HỆ THỐNG NHÚNG
(Embedded System)**

TP.HCM-2024

NỘI DUNG MÔN HỌC

Chương 1: GIỚI THIỆU HỆ THỐNG NHÚNG

Chương 2: KỸ THUẬT GIAO TIẾP NGOẠI VI

Chương 3: CẤU TRÚC PHẦN MỀM

Chương 4: THIẾT KẾ HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

1. HỆ THỐNG NHÚNG
2. PHẦN CỨNG HỆ THỐNG NHÚNG
3. PHẦN MỀM HỆ THỐNG NHÚNG
4. CÔNG NGHỆ Multi-Core, SoC, and FPGA
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2. PHẦN CỨNG HỆ THỐNG NHÚNG

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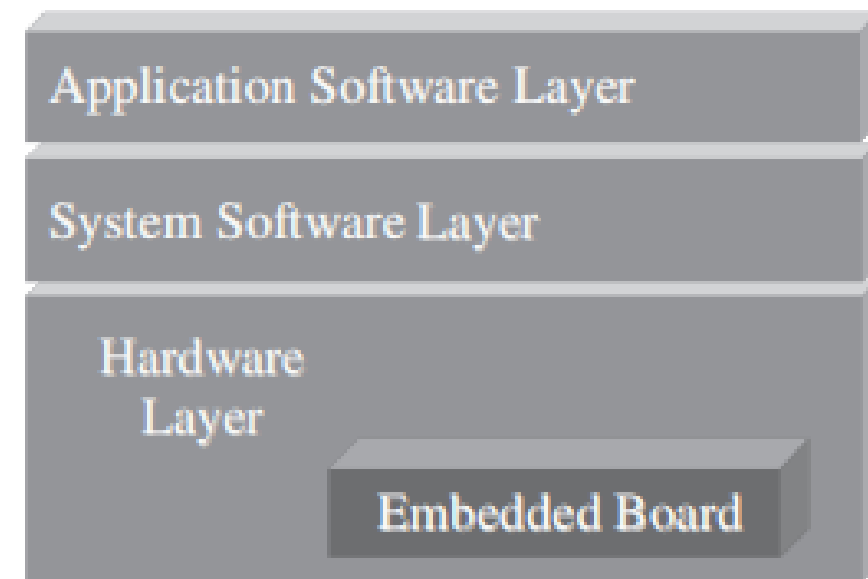
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2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.1 The Embedded Board and the von Neumann Model

- Central Processing Unit (CPU) – *master processor*
- Memory
- Input Device(s) – *input slave processors and relative electrical components*
- Output Device(s) – *output slave processors and relative electrical components*
- Data Pathway(s)/Bus(es) – *interconnects, providing a “highway” for data to travel on from one component to another: wires, bus bridges, and/or bus controllers*



(Nguồn: Internet)

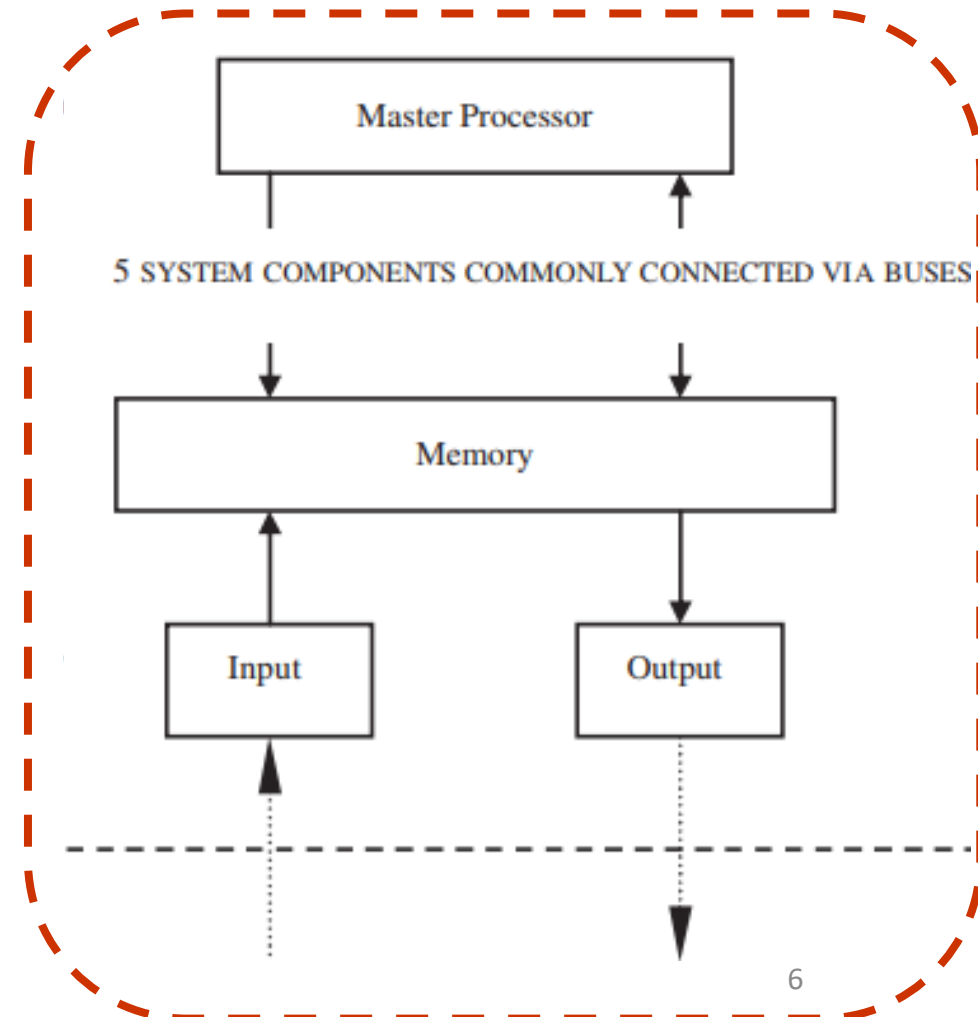
2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.1 The Embedded Board and the von Neumann Model

+ **Five categories** are based upon the major elements defined by the **von Neumann model** (or *Princeton model*) as Figure, is a result of the published work of **John von Neumann**.

+ **Von Neumann model** is an electronic device's *hardware architecture*, which can be applied as a means of understanding **embedded systems hardware**.



2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

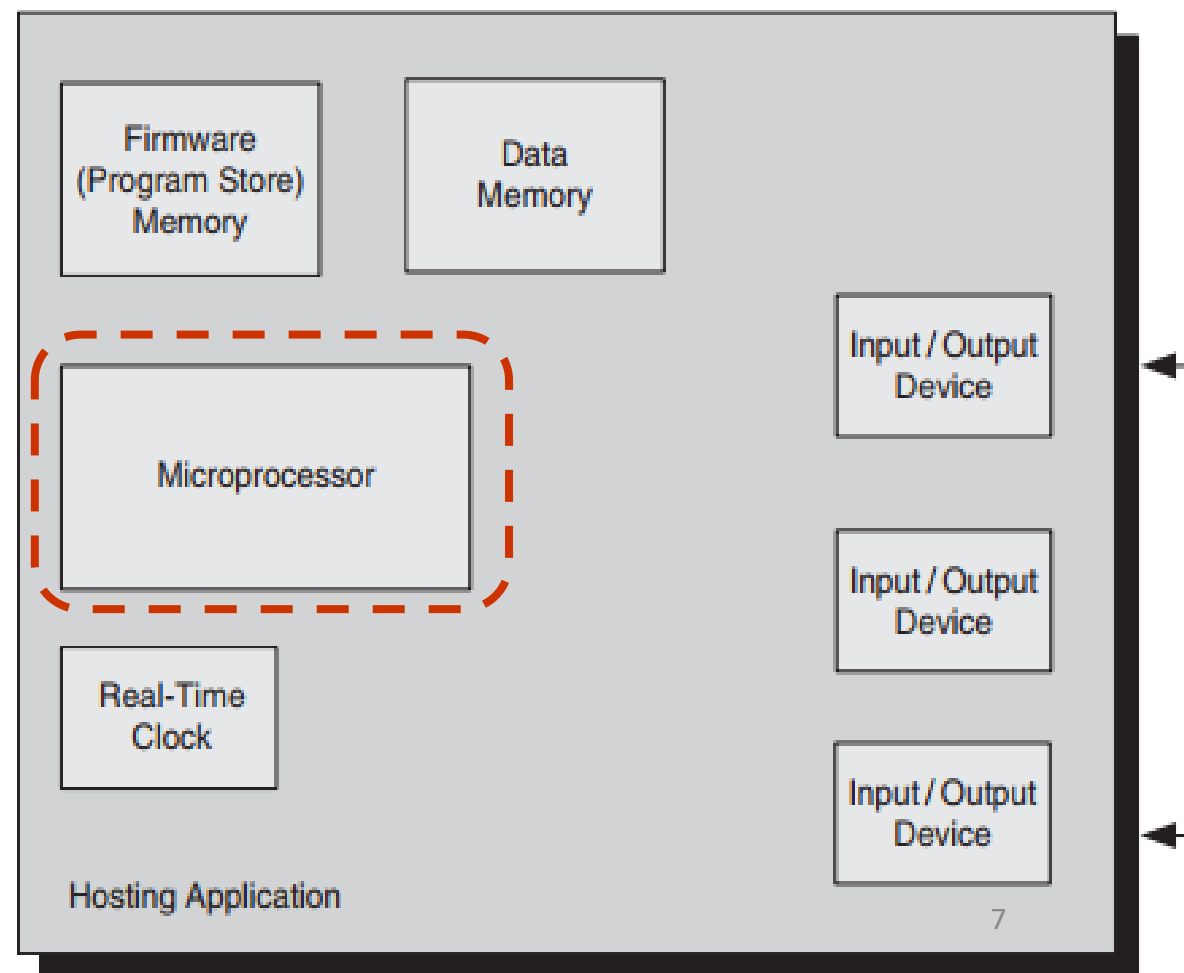
2.1 The Embedded Board and the von Neumann Model

A. Micro-processor

+ A *microprocessor* is an integrated implementation of the CPU portion (*control and arithmetic and logical unit*) of the machine.

+ To implement a complete (embedded) computer system, we must still include the *input/output subsystems* and the *external* (to the *microprocessor*) *memory system*.

(Nguồn: Internet)



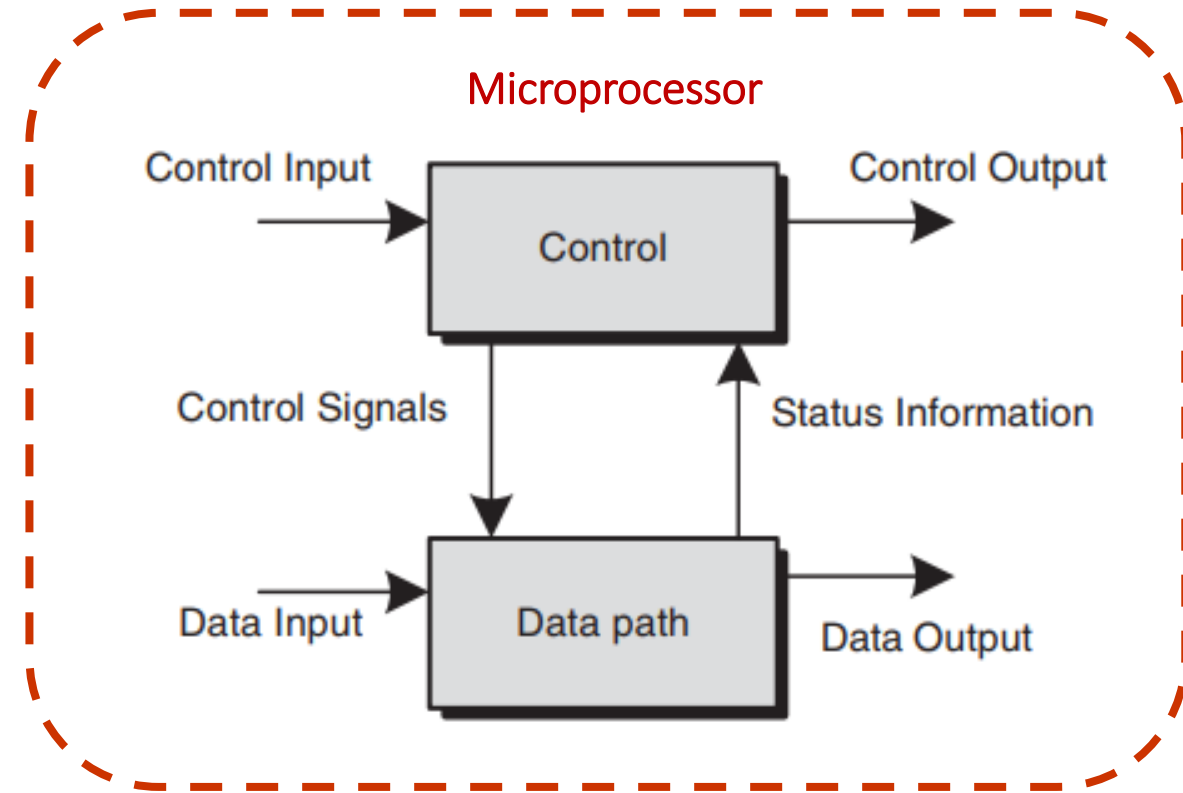
2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.1 The Embedded Board and the von Neumann Model

A. Micro-processor

+ **Datapath**: is a collection of registers and an associated set of *micro-operations* on the data held in the registers.



A Control and Datapath block Diagram

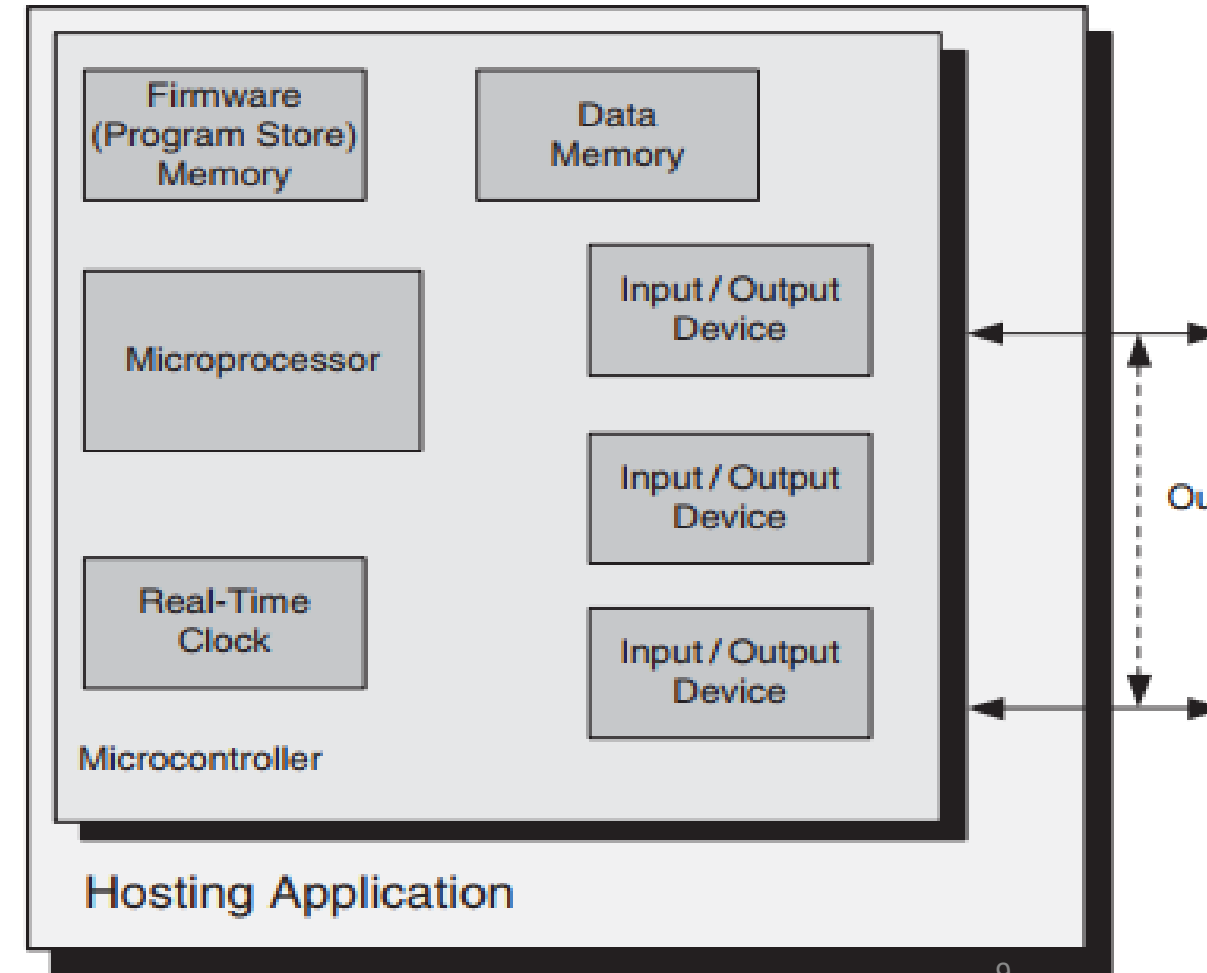
2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.1 The Embedded Board and the von Neumann Model

B. Micro-controller

- + The *Microcontroller* brings together the microprocessor **core** and a rich collection of peripherals and I/O capability into a single integrated circuit.
- + *Microcontrollers* find great utility in basic embedded applications where **low cost** is a significant constraint.



2. PHẦN CỨNG HỆ THỐNG NHÚNG

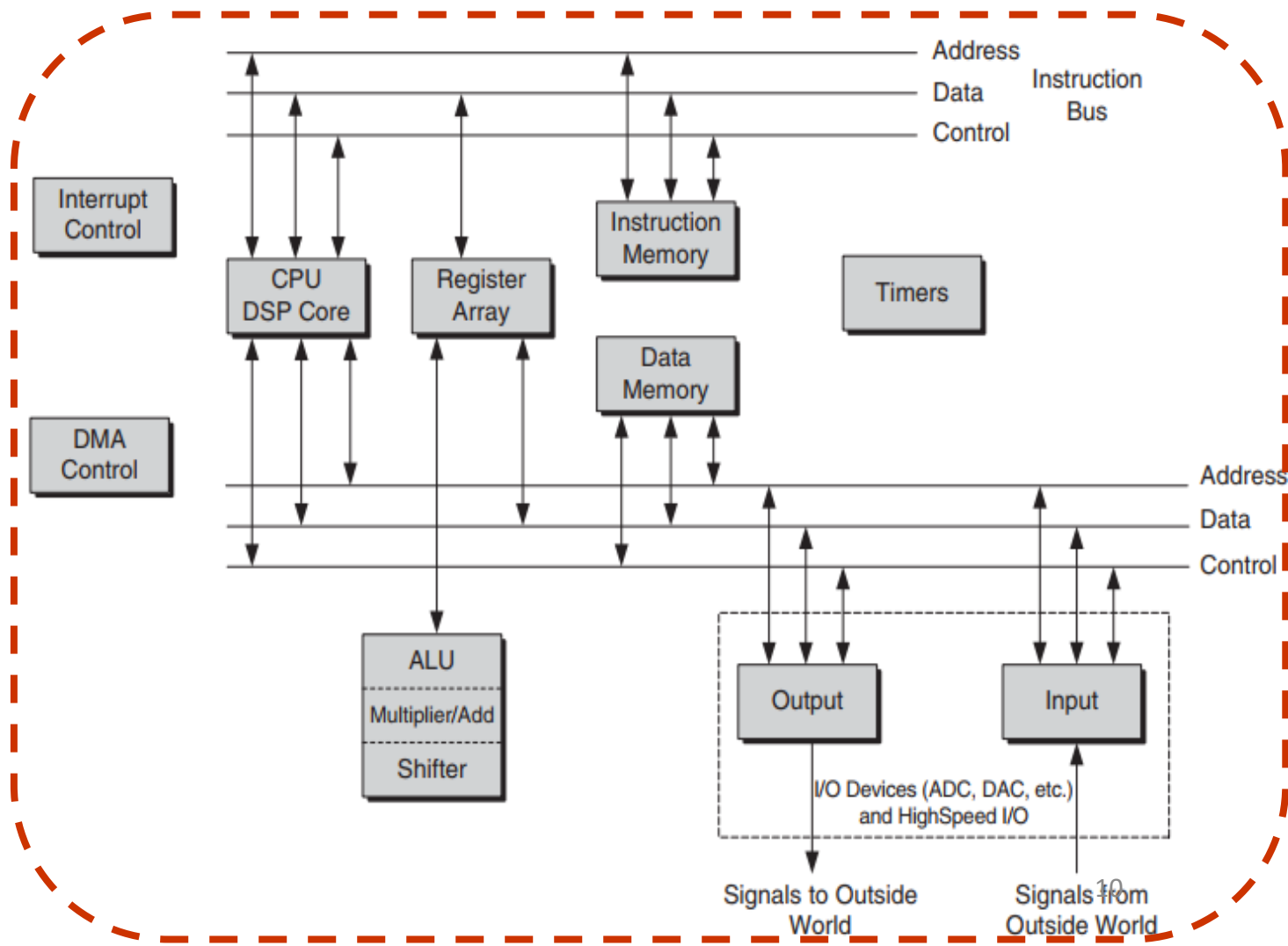
CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.1 The Embedded Board and the von Neumann Model

C. Digital Signal Processor (DSP)

+ The **DSP** is typically used in conjunction with a **general purpose processor** to perform specialized tasks such as *image, speech, audio, video processing*.

+ The **DSP** is architected as a **Harvard** rather than the **von Neumann** machine.



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

- + **Processors** are the main functional units of an embedded board, acting as the **central controlling device**, and can have additional slave processors.
- + **Slave processors** may either extend the instruction set of the master processor or act to **manage memory, buses, and I/O (input/output) devices**.
- + **Why Use an Integrated Processor?**

Integrated processors simplify the entire **board design** and the **power requirements** of components integrated into a chip → provide a *smaller and cheaper* embedded board.

2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.2 The Emt

Architecture	Processor	Manufacturer
AMD	Au1xxx	Advanced Micro Devices, ...
ARM	ARM7, ARM9, ...	ARM, ...
C16X	C167CS, C165H, C164CI, ...	Infineon, ...
ColdFire	5282, 5272, 5307, 5407, ...	Motorola/Freescale, ...
I960	I960	Vmetro, ...
M32/R	32170, 32180, 32182, 32192, ...	Renesas/Mitsubishi, ...
M Core	MMC2113, MMC2114, ...	Motorola/Freescale
MIPS32	R3K, R4K, 5K, 16, ...	MTI4kx, IDT, MIPS Technologies, ...
NEC	Vr55xx, Vr54xx, Vr41xx	NEC Corporation, ...
PowerPC	82xx, 74xx, 8xx, 7xx, 6xx, 5xx, 4xx	IBM, Motorola/Freescale, ...
68k	680x0 (68K, 68030, 68040, 68060, ...), 683xx	Motorola/Freescale, ...
SuperH (SH)	SH3 (7702, 7707, 7708, 7709), SH4 (7750)	Hitachi, ...
SHARC	SHARC	Analog Devices, Transtech DSP, Radstone, ...
strongARM	strongARM	Intel, ...
SPARC	UltraSPARC II	Sun Microsystems, ...
TMS320C6xxx	TMS320C6xxx	Texas Instruments, ...
x86	X86 [386, 486, Pentium (II, III, IV)...]	Intel, Transmeta, National Semiconductor, Atlas, ...
TriCore	TriCore1, TriCore2, ...	Infineon, ...

2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.1. Some ISA Architecture models of Processor

- + The **features** that are built into an **Architecture's Instruction Set** are commonly referred to as the **Instruction Set Architecture (ISA)**.
- + The **features** or **operations** used by programmers to create programs for that architecture: *storage, addressing modes, interrupts handling*.
- + The **ISA implementation** is a determining factor in defining important characteristics of an embedded design: *performance, design time, available functionality, and cost*.
- It serves as an interface **to allow easy communication** between the **programmer** and **the hardware**. It prepares the processor to respond to all the user commands.

2. PHẦN CỨNG HỆ THỐNG NHÚNG

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2.2 The Embedded Processors

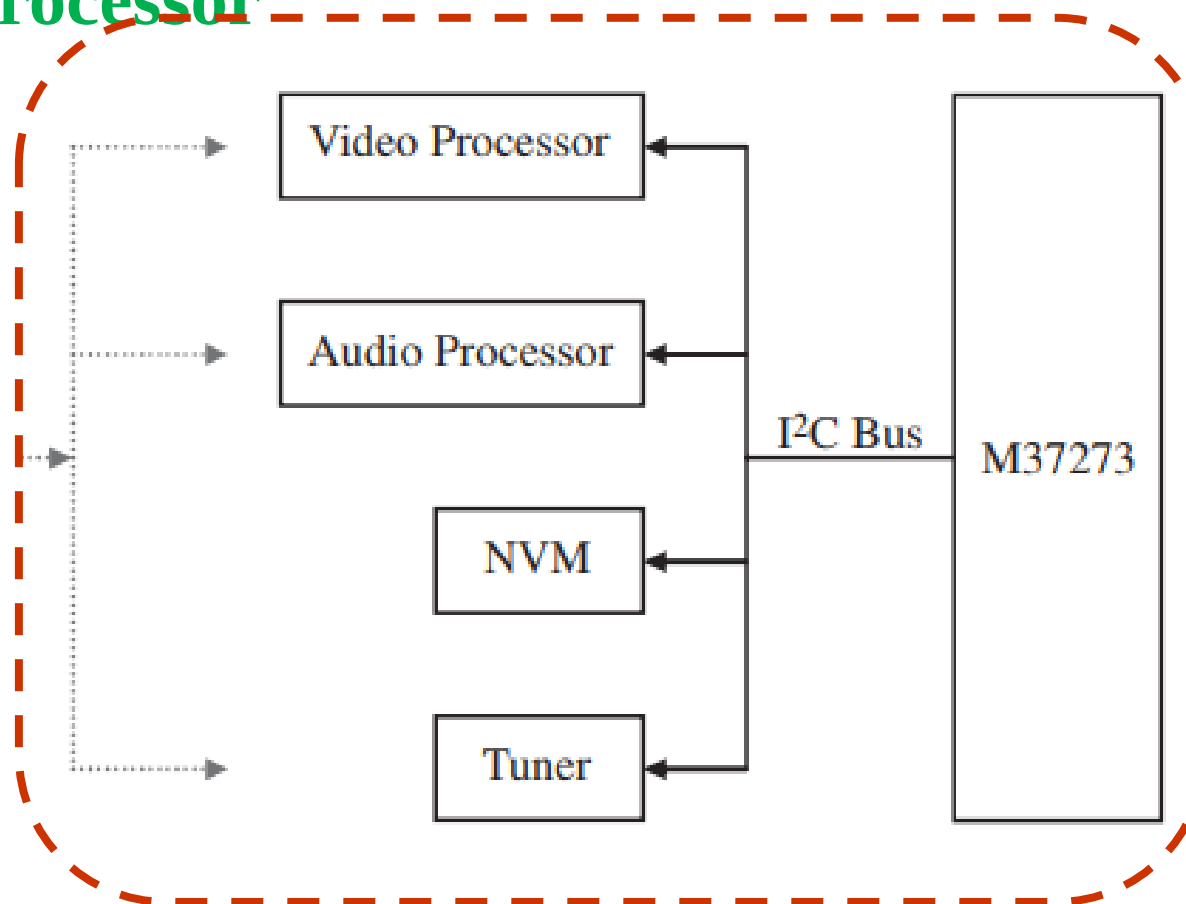
2.2.1. Some ISA Architecture models of Processor

A. Application-Specific ISA Models

→ define processors that are intended for specific embedded applications.

1) Controller Model: is implemented in processors that **are not required** to perform complex data manipulation.

(for example, *video* and *audio processors* that are used as *slave processors* on a TV board)



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.1. ISA Architecture Models

A. Application-Specific ISA models

2) **Datapath model**: is implemented in processors whose purpose is to repeatedly perform fixed computations on different sets of data.

(a *digital signal processors (DSPs) in digital cellphone*).

2. PHẦN CỨNG HỆ THỐNG NHÚNG

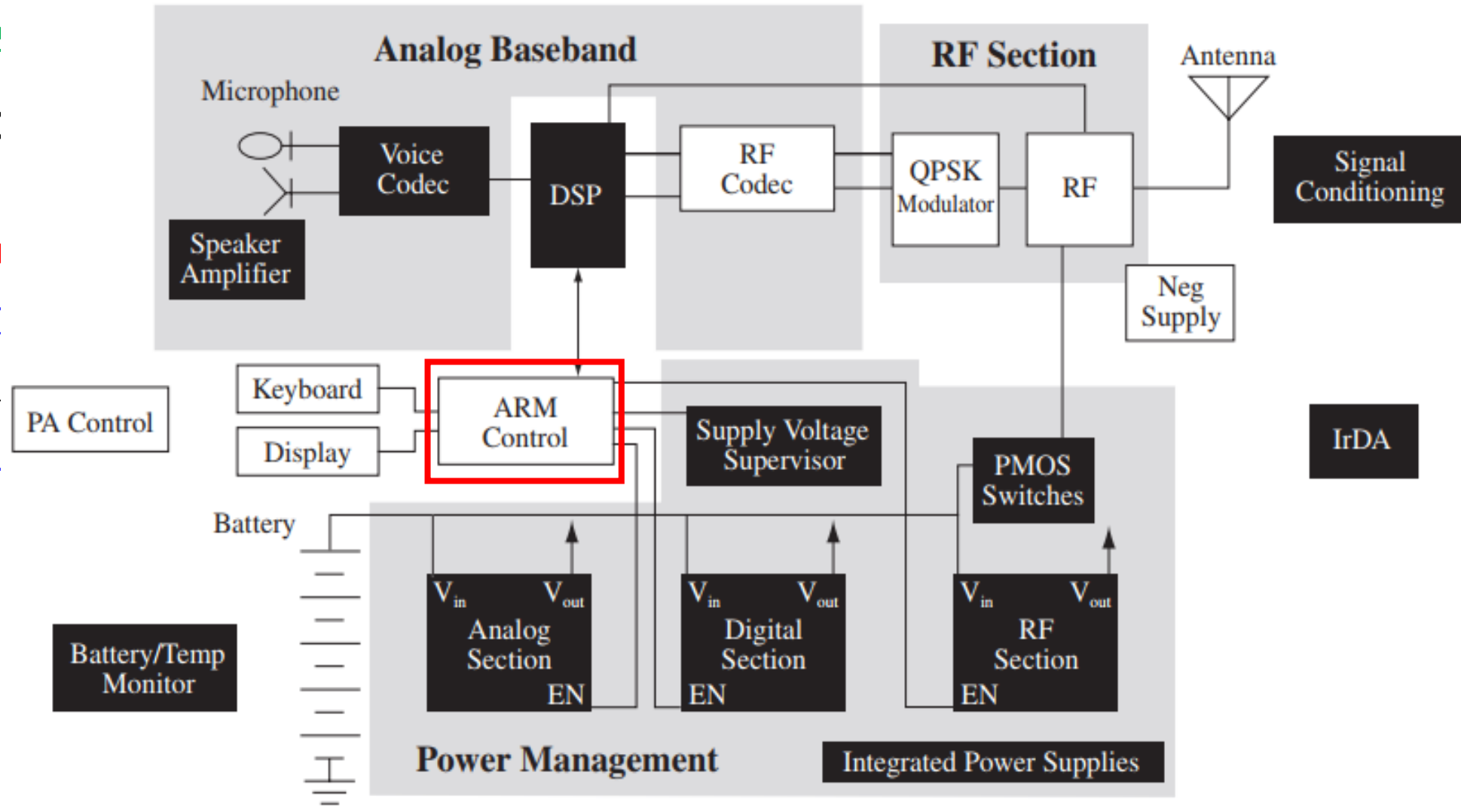
CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.1. ISA Archite

A. Application-Sp

2) Datapath mode
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(a *digital signal pr*



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.1. ISA Architecture Models

B. General-Purpose ISA models → are typically implemented in processors targeted to be used in a wide variety of embedded systems.

+ **Complex Instruction Set Computing (CISC)**: The **CISC ISA** defines complex operations made up of several instructions.

+ **Reduced Instruction Set Computing (RISC)**: The **RISC ISA** defines:

- *RISC with simpler and/or fewer operations made up of fewer instructions;*
- *RISC has a reduced number of cycles per available operation.*

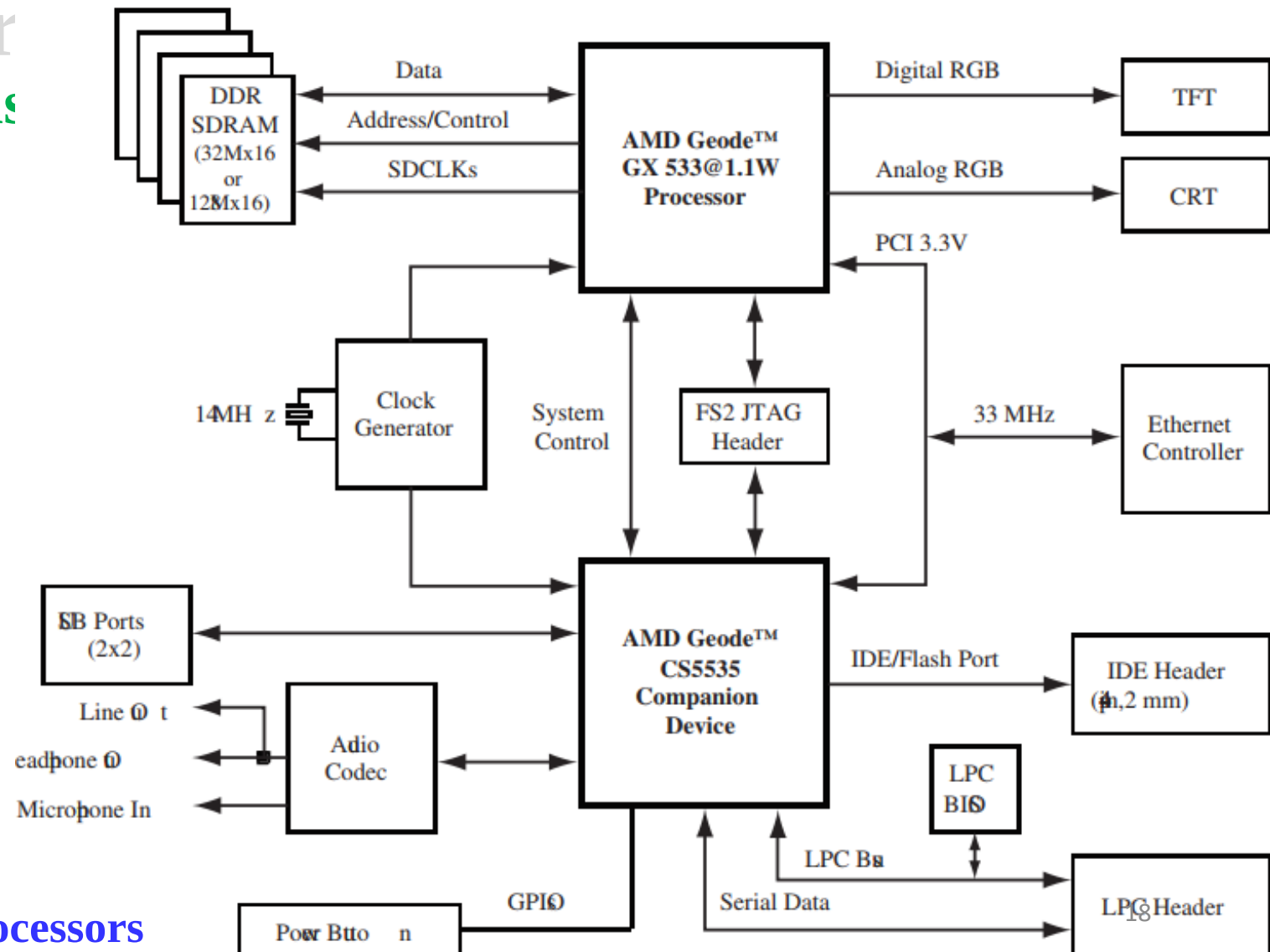
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2.2 The Embedded Pr

2.2.1. ISA Architecture Models

+ **CISC** model (as **Figure**)



CISC processors

2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

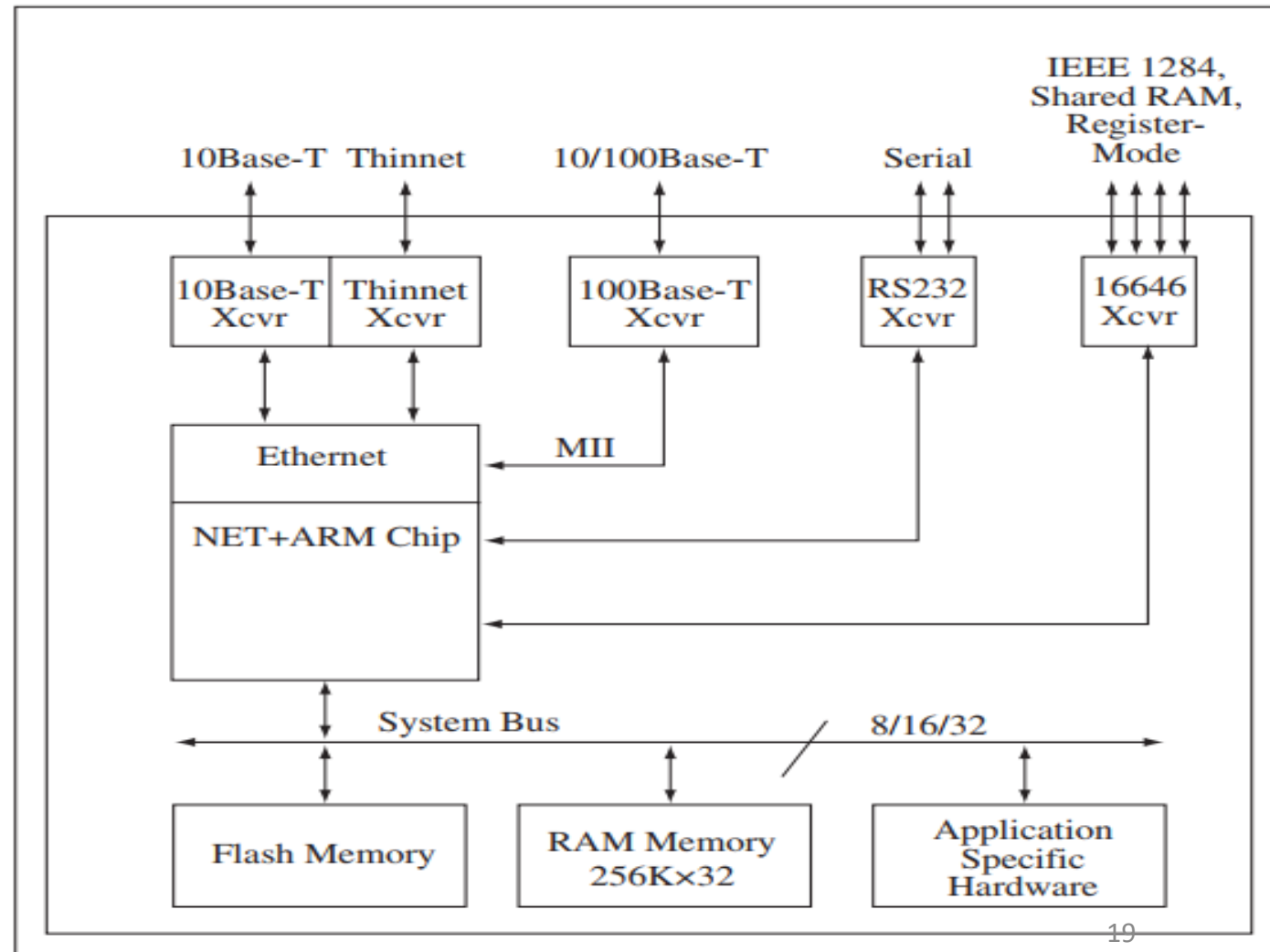
2.2 The Embedded Processors

2.2.1. ISA Architecture Models

+ **RISC processors** have only *one-cycle operations*, whereas **CISCs** typically have *multiple cycle operations*.

→ **RISC processors** have become *more complex*, while **CISC** processors have become *more efficient* to compete with their RISC counterparts.

RISC processors



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.1. ISA Architecture Models

C. Instruction-Level Parallelism ISA models

→ are similar to general-purpose ISAs, except that they execute multiple instructions *in parallel*.

→ are considered higher evolutions of the **RISC ISA**, which typically has *one-cycle operations*.

1) **Single Instruction Multiple Data model (SIMD)**

- **SIMD** is designed to **process an instruction** simultaneously *on multiple data components* that *require action* to be performed on them.

2. PHẦN CỨNG HỆ THỐNG NHÚNG

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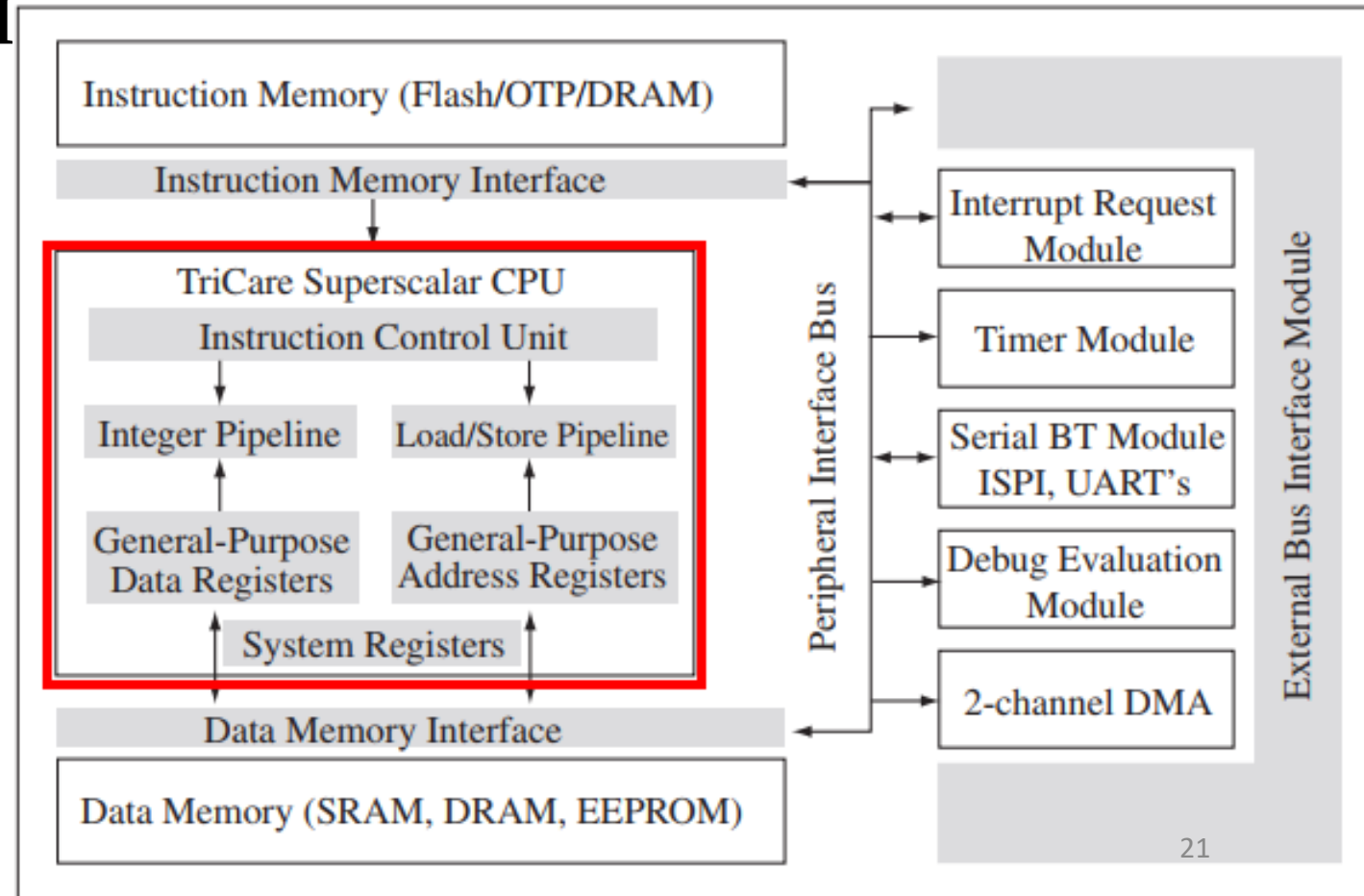
2.2 The Embedded Processors

2.2.1. ISA Architecture Models

C. Instruction-Level Parallelism I

2) Superscalar Machine model

→ process **multiple instructions** simultaneously **within one clock cycle** through the implementation of *multiple functional components* within the processor.



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.1. ISA Architecture Models

C. Instruction-Level Parallelism ISA models

3) Very Long Instruction Word Computing model (VLIW)

- **VLIW**, in which a **very long instruction word** is made up of multiple operations.
- These operations processed **in parallel** by multiple execution units within the processor .

2. PHẦN CỨNG HỆ THỐNG NHÚNG

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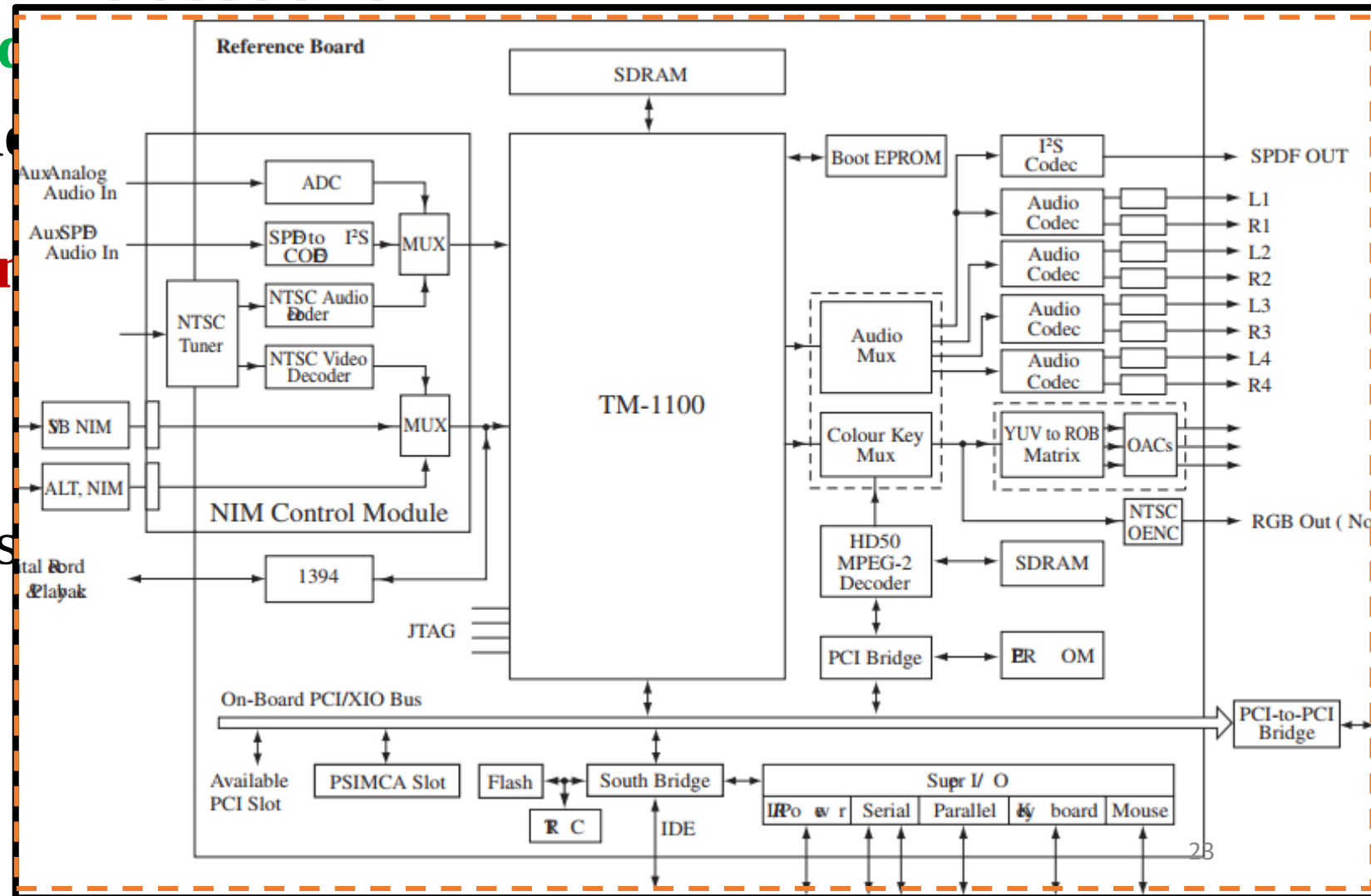
2.2 The Embedded Processors

2.2.1. ISA Architecture Model

C. Instruction-Level Parallel

3) Very Long Instruction

- In which a very long operations.
- These operations process within the processor .



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.2. Internal Processor Design

ISA's components in a processor:

- CPU, memory, input components, output components and buses.

+ As mentioned, these components are the basis of the **von Neumann**.

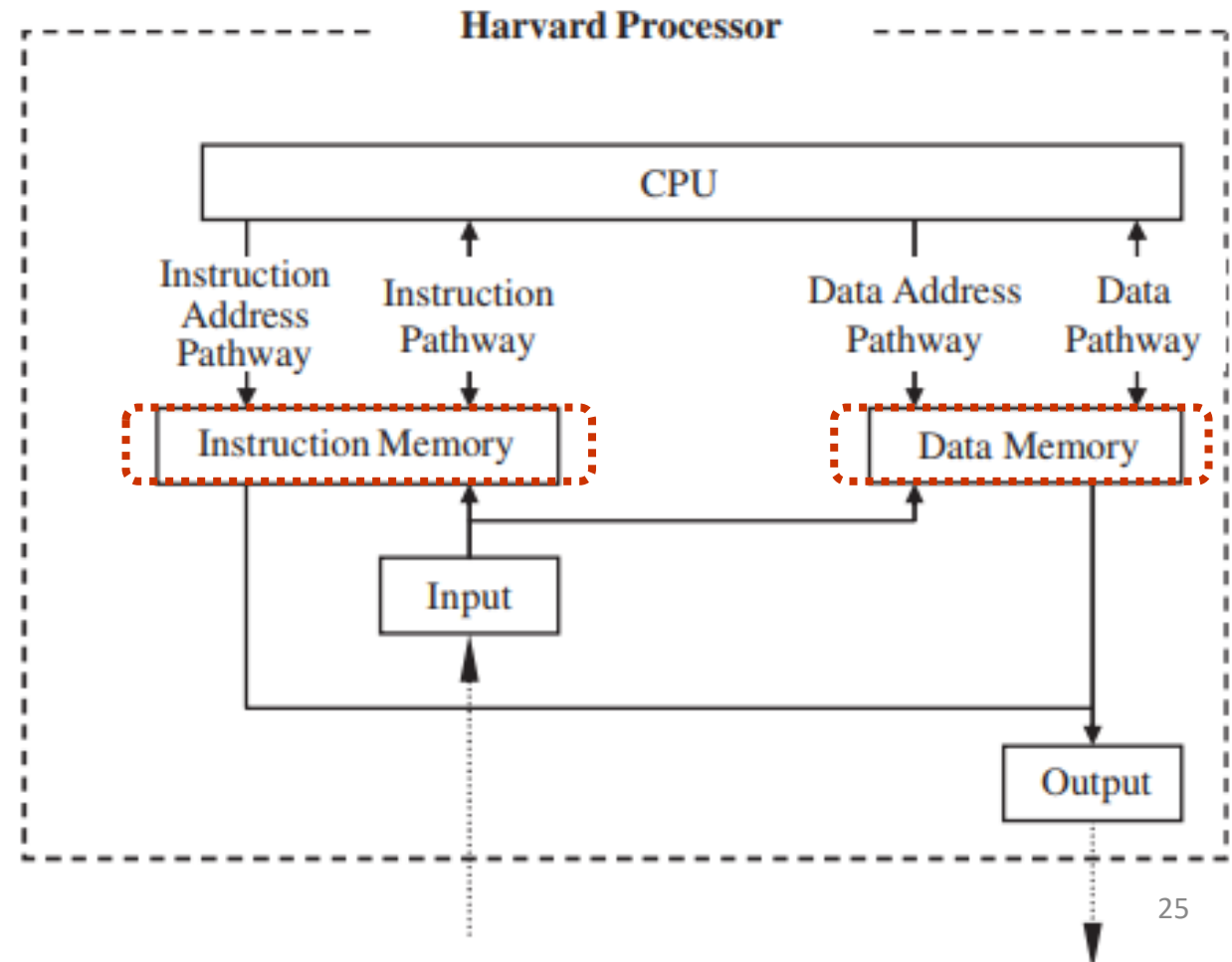
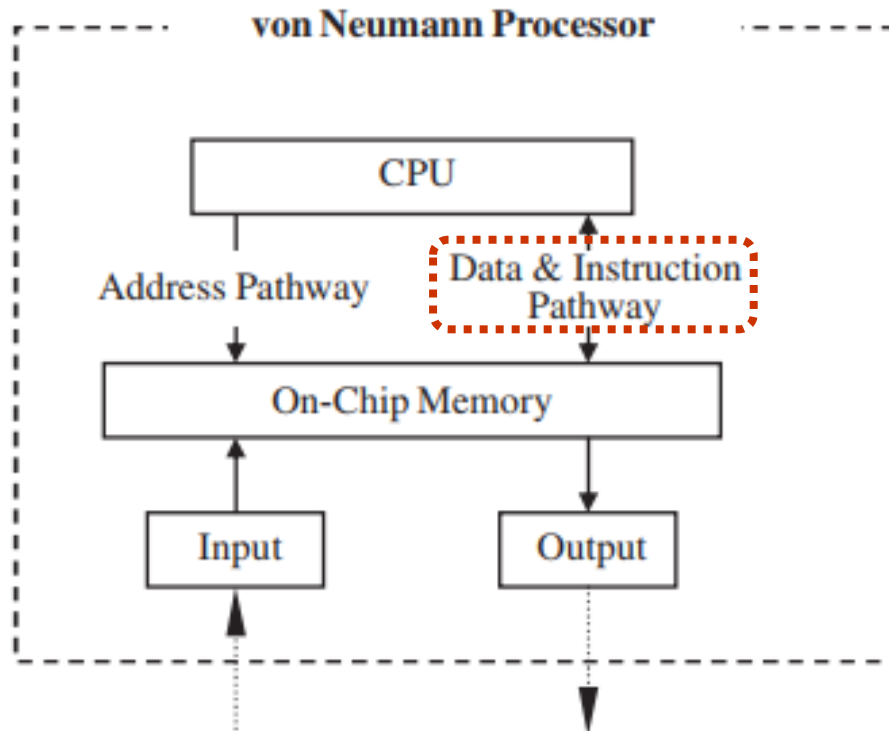
+ **Harvard** architecture defines separate memory spaces for instructions and data; *separate data and instruction buses* allow for *simultaneous fetches and transfers* to occur.

2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.2. Internal Processor Design



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.2. Internal Processor Design

❑ Central Processing Unit (CPU)

→ The instruction cycle comprises *3 fundamental operations*:

- *fetch*: retrieve an instruction from memory.
- *decode*: is performed when the op-code field in the instruction is extracted from the instruction and decoded by the Instruction Decoder.
- *execute*: based on the value contained in the op-code field, the control logic performs the sequence of steps necessary to execute the instruction.

→ This **three-step process** is commonly referred to as a **three-stage pipeline**, and most recent CPUs are pipelined designs.

2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

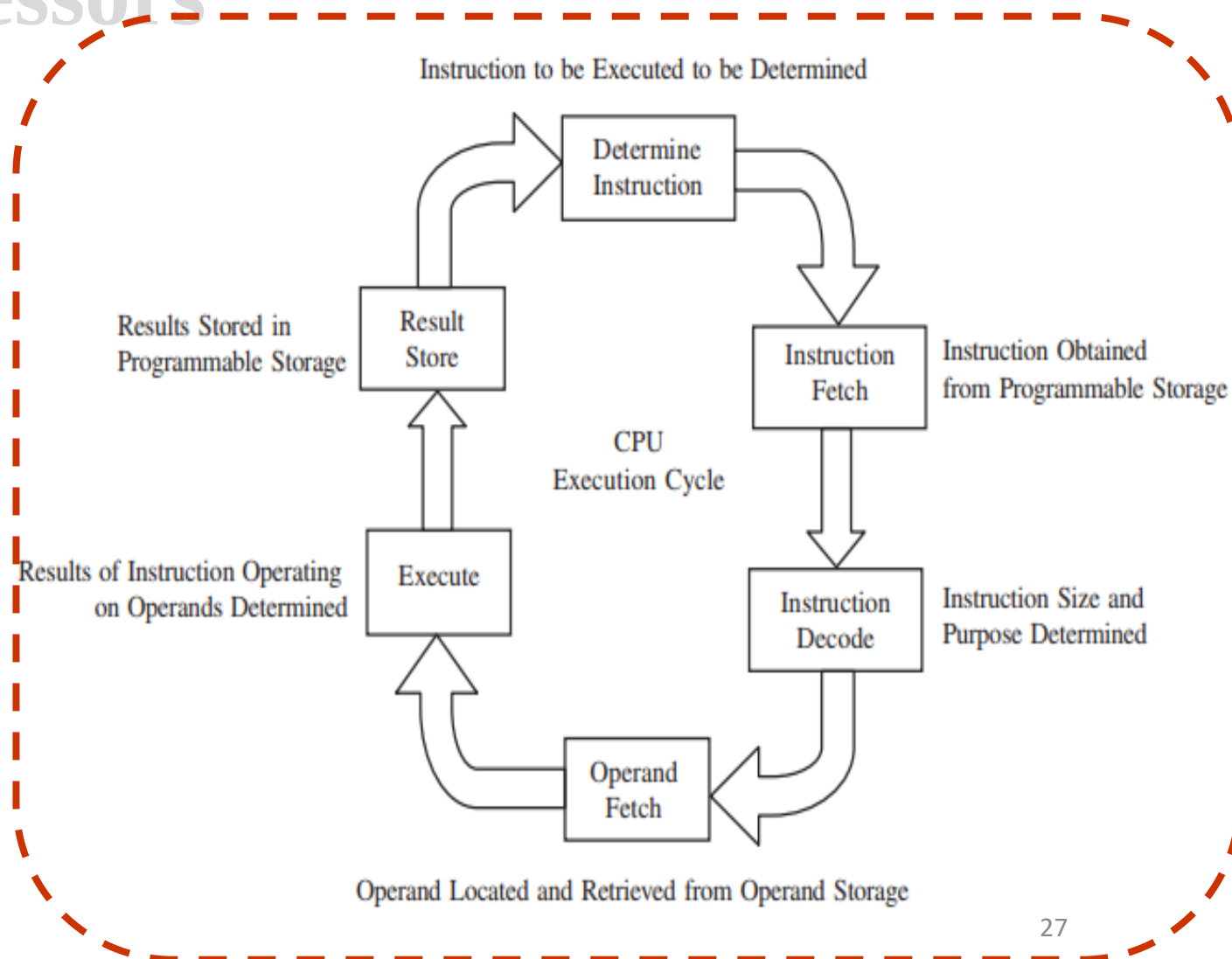
2.2 The Embedded Processors

2.2.2. Internal Processor Design

❑ Central Processing Unit (CPU)

→ Figure executes the cycle of **fetching**, **decoding**, and **executing** instructions.

→ This three-step process is commonly referred to as a **three-stage pipeline**, and most recent CPUs are pipelined designs.



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.2. Internal Processor Design

❑ Central Processing Unit (CPU)

→ As defined by the **von Neumann** model, this instruction cycle is implemented through some combination of 4 major CPU components:

- **Arithmetic Logic Unit (ALU)** – implements the ISA's operations
- **Registers** – a type of fast memory.
- **Control Unit (CU)** – manages the entire **fetching, decodeing, execution** cycle.
- **Internal CPU buses** – interconnect the ALU, Registers, and the CU.

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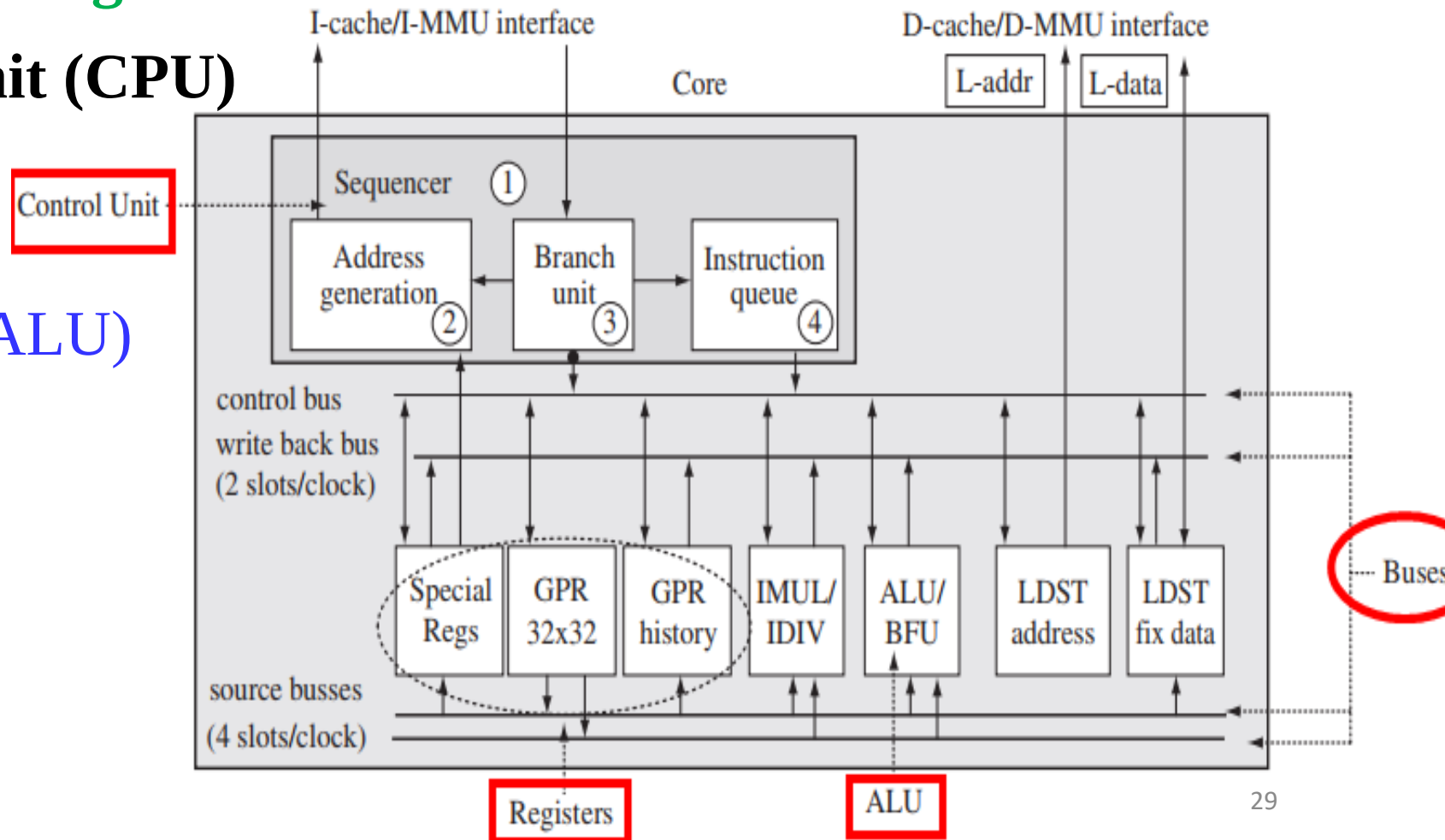
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2.2 The Embedded Processors

2.2.2. Internal Processor Design

❑ Central Processing Unit (CPU)

- + Arithmetic Logic Unit (ALU)
- + Registers
- + Control Unit (CU)
- + Internal CPU buses



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.3. Processor Performance

- Measuring **processor performance** based upon the *processor's behavior* over a given *length of time*. The one most common definitions (**throughput**), the *amount of work* the CPU completes in a given period of time.
- **Oscillator**: produces clock pulse signal running at a constant number of clock cycles per second, or **clock rate**, is expressed in MHz (megahertz).
 - **Clock period** (**cycle time**): the length of time a CPU takes to complete a clock cycle (is the inverse of clock rate, **$1/\text{clock rate}$**).

2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2.3. Processor Performance

- **CPI** (*average number of Clock cycles Per Instruction*)

$$CPI_{(MHz)} = \sum (CPI_{per\ instruction} * instruction\ frequency)$$

- **CPU's execution time**: is the total time the processor takes to process some program in seconds per program.

$$CPU\ execution\ time\ (s) = instruction\ count * (CPI_{per\ instruction} / clock\ rate)$$

- **CPU throughput** (or **bandwidth**): reflects the amount of work the CPU does in a period of time.

$$CPU\ throughput = 1 / CPU\ execution\ time(s) = CPU\ performance(bytes/s, MB/s)$$

2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.2 The Embedded Processors

2.2.3. Processor Performance

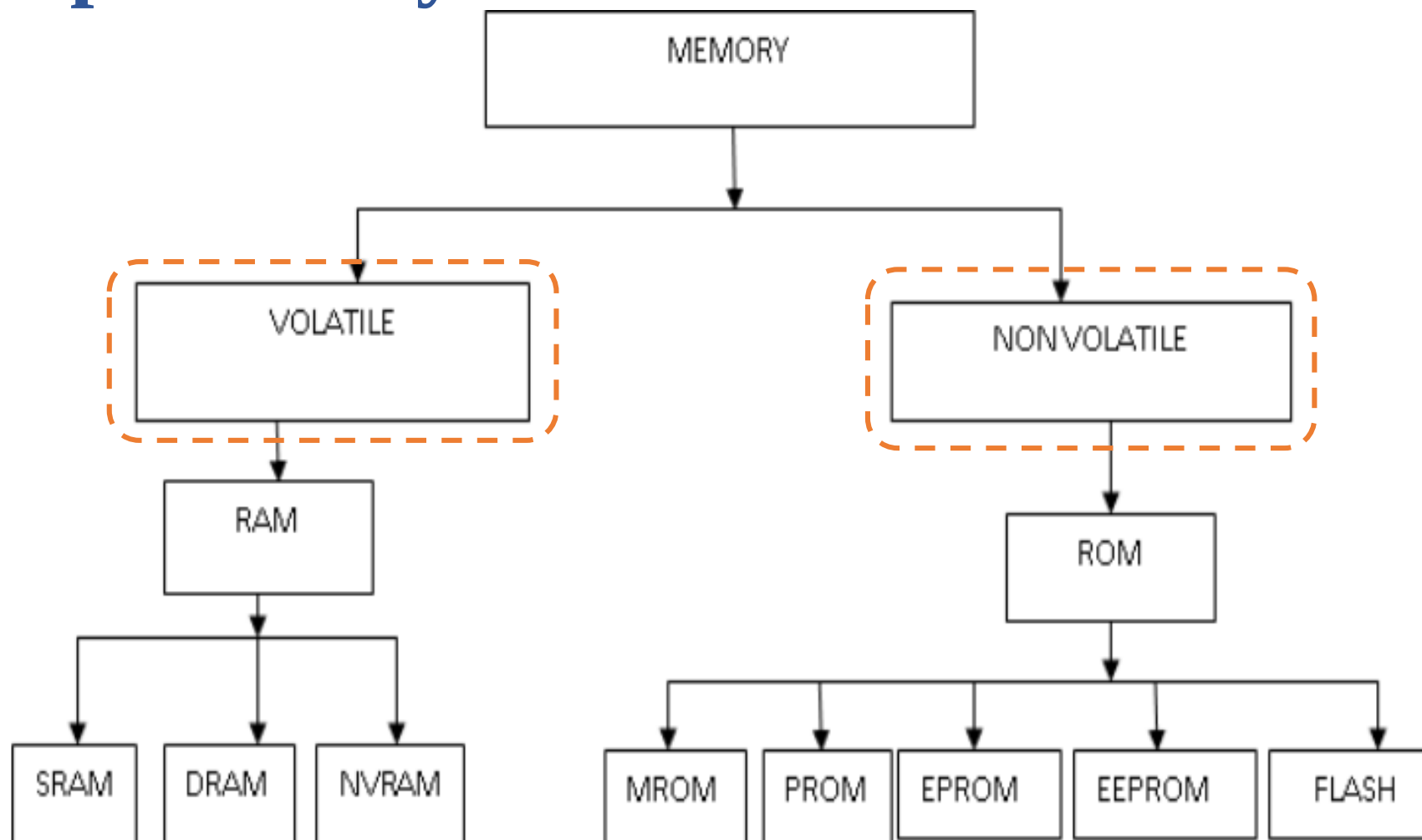
Similar to *throughput*, other definitions of *Performance* as below:

- **Processor's Responsiveness (*latency*):** the *length of elapsed time* a processor takes to respond to some events.
- **Processor's Availability:** the *amount of time* the processor runs normally *without failure*.
- **Reliability** (or MTBF=*mean time* between failures).
- **Recoverability** (or MTTR=*mean time* to recover).

2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.3 On-Chip Memory



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.3 On-Chip Memory

- + Generally, both primary and secondary memory can be divided into two groups, ***non-volatile*** or ***volatile***. **Non-volatile memory** is memory that can store data after the main power source to the board has been shut off.
- + **On embedded boards**, there are:
 - **Non-volatile**: **Read-Only Memory (ROM)** and **auxiliary** memory (*magnetic disks, magnetic tapes, and optical discs*).
 - **Volatile**: **Random-Access Memory (RAM)**.

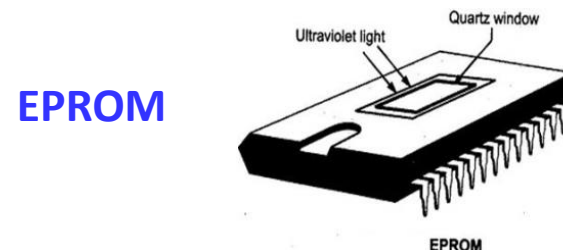
2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.3 On-Chip Memory

A. Read-Only Memory (ROM)

- + **MR**OM (*mask ROM*): that is **permanently etched** into the microchip during the manufacturing of the processor, and cannot be modified later.
- + **P**ROM (*programmable ROM*), (*one-time programmables- OTPs*): that can be integrated on-chip, that is one-time programmable by programmer.
- + **EP**ROM (*erasable programmable ROM*): can be erased and reprogrammed more than once (*depends on the processor*), is erased by **intense ultraviolet light**.
- + **EEP**ROM (*electrically erasable programmable ROM*): is cheaper and faster variation of the EEPROM is **Flash memory**.



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.3 On-Chip Memory

B. Random-Access Memory (RAM)

+ **DRAM (dynamic RAM)**: using a **stored charge mechanism**, which can leak away, hence DRAM needs to be **refreshed periodically**. It is used for larger memory systems.

→ This makes DRAM more complex and power-consuming. However, as this form of memory is **less expensive** to produce compared to SRAM.

+ **SRAM (static RAM)**: is designed with bit storage implemented using a **latch-type mechanism** (stored data **does not** have to be **refreshed**).

→ It is used for **higher speed** memory systems because they are faster than DRAM designs. It is capable of running at **133 MHz**, a typical clock frequency.

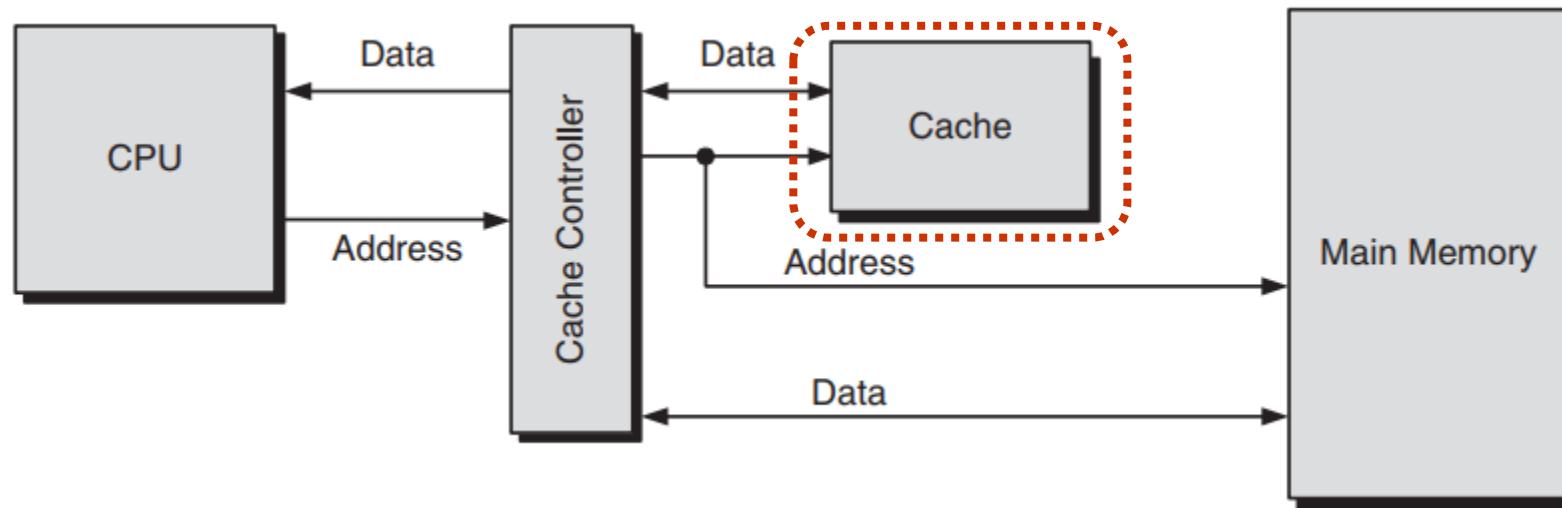
2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.3 On-Chip Memory

C. Cache Memory

- + When program execution needs **an instruction** or **data** that is not in the cache, **it must be** brought in from main memory (typically using **DRAM**).
- + **The goal** is to operate out of cache memory– to the **greatest extent possible**. Caching requires a certain amount of higher speed memory (typically **SRAM**).



2. PHẦN CỨNG HỆ THỐNG NHÚNG

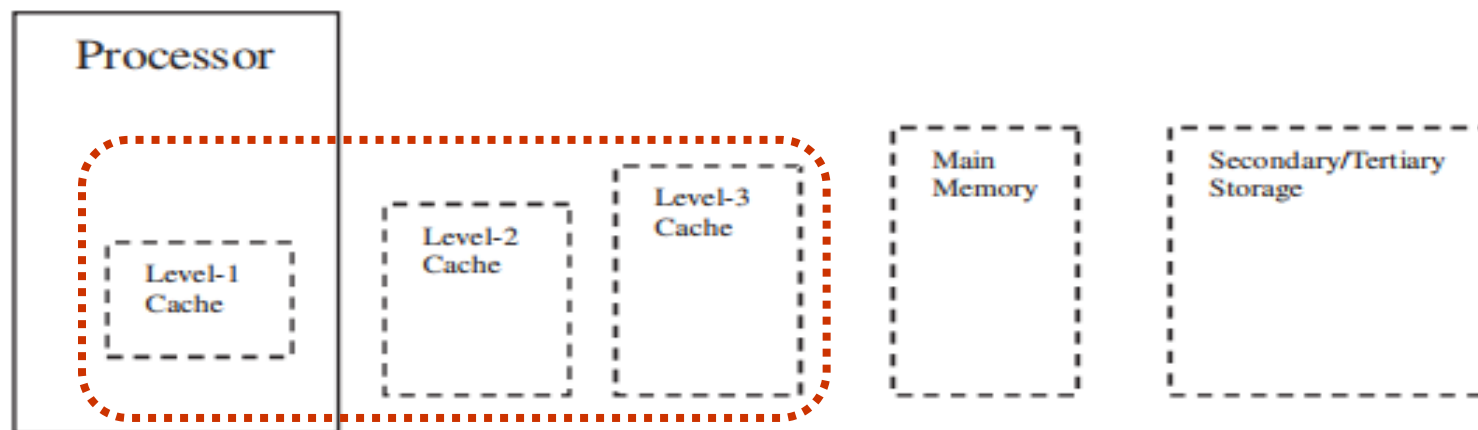
CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.3 On-Chip Memory

C. Cache Memory

+ CPUs have multilevel cache memory. They are referred to as L1 for level one, L2 for level two, L3 for level three, ... **A CPU directly works with L1 cache.**

+ **The goal** of the **L1** cache is to line up with the CPU so that it has data to work on every processor cycle **at the best**. It is a **small, high-speed cache** incorporated right onto the processor's chip.



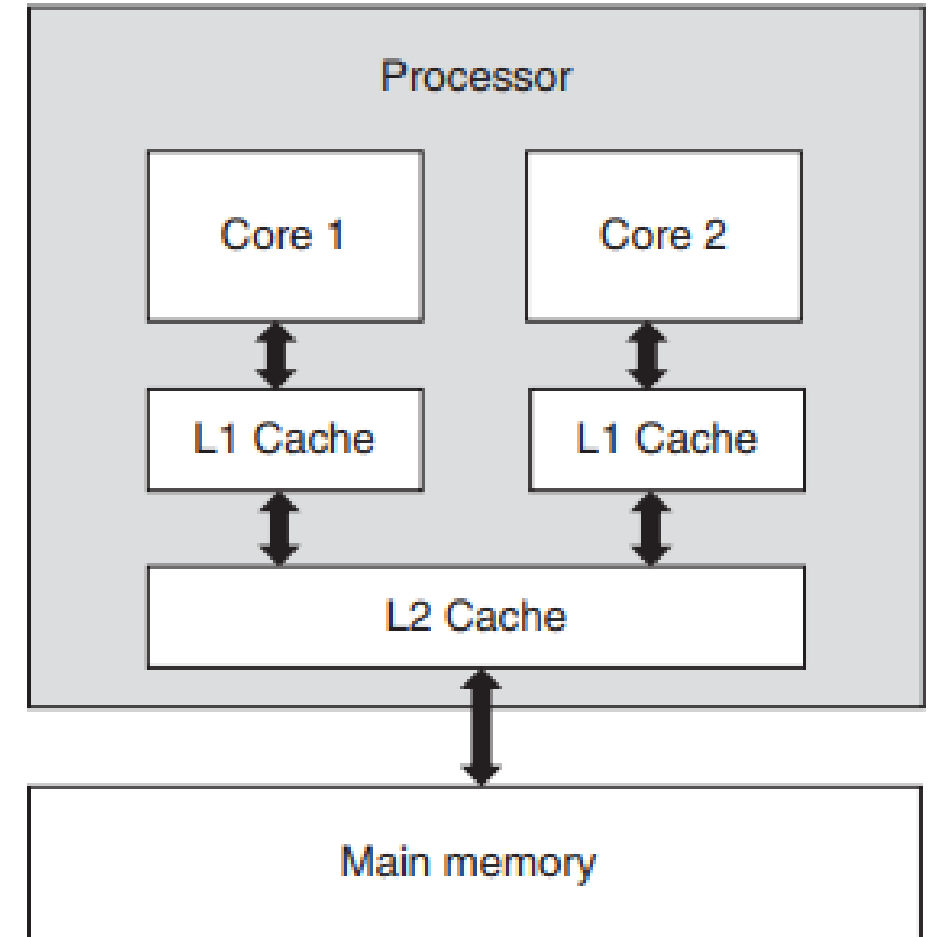
2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.3 On-Chip Memory

C. Cache Memory

- + The **L1** cache typically ranges in size from 8 to 64 KB and uses the high-speed SRAM.
- + The **L2** cache is expected to feed data to the **L1** cache every few processor cycles, while the **L3** cache can feed data to the **L2** cache at a further slower rate.
- + The three levels exchange data based on CPU requirements.



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.4 Board Input/output (I/O)

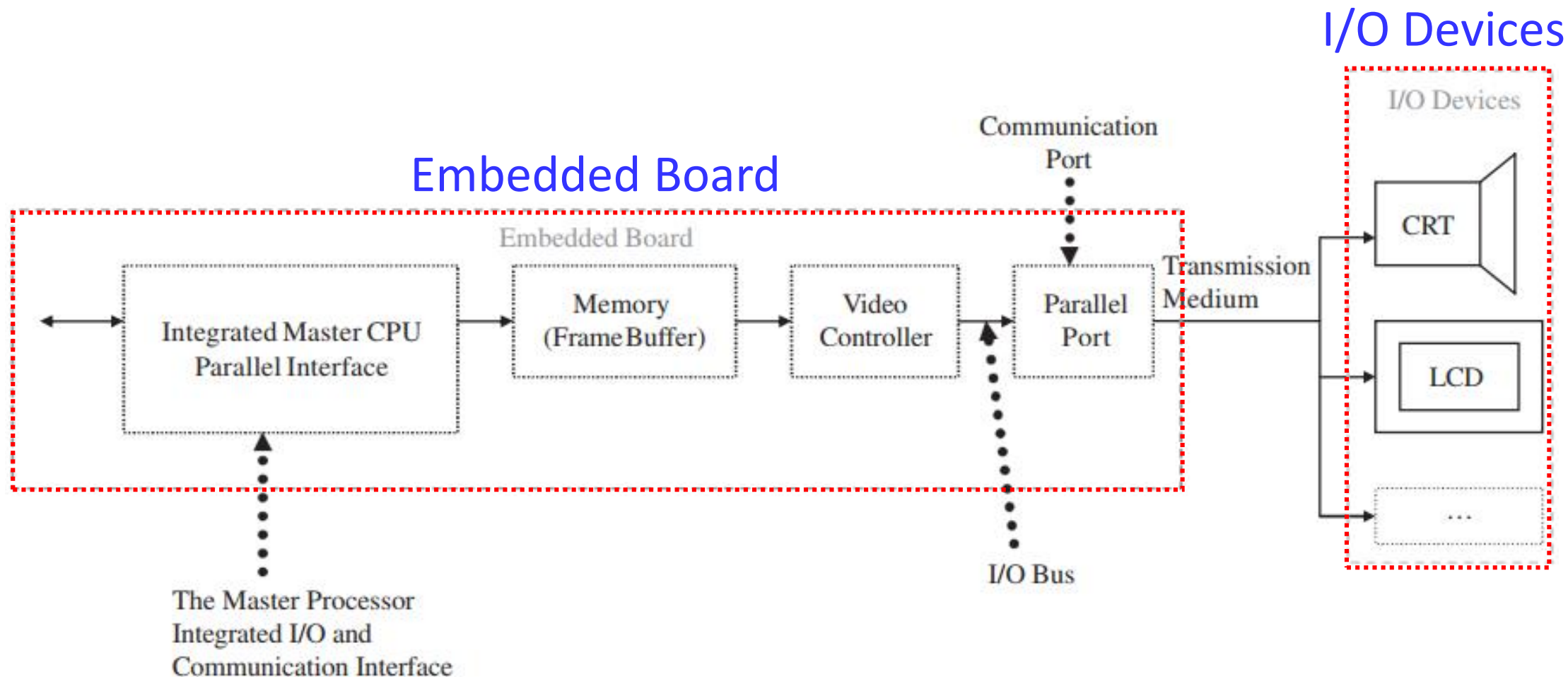
❑ Processor I/O components

- + Networking and communications I/O.
- + Input (keyboard, mouse, remote control, voice, etc.).
- + Graphics and output I/O (touch screen, CRT, printers, LEDs, etc.).
- + Storage I/O (optical disk controllers, magnetic disk controllers, magnetic tape controllers, etc.).
- + Debugging I/O (BDM, JTAG, serial port, parallel port, etc.).
- + Real-time and miscellaneous I/O (Timers/counters, ADC and DAC, key, switches, and so on).

2. PHẦN CỨNG HỆ THỐNG NHÚNG

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

2.4 Board Input/output (I/O)



2. PHẦN CỨNG HỆ THỐNG NHÚNG

2.5 Board Buses

- + A **bus's performance** is typically measured by its **bandwidth**, the amount of data a bus can transfer for a given length of time.
- + The actual physical design of the bus (its *length*, the *number of lines*, the *number of supported devices*, and so on) **limits or enhances** its performance.
- + The shorter the bus, the fewer connected devices, and the more data lines, typically the faster the bus and the higher its bandwidth.

CHƯƠNG 1: GIỚI THIỆU HỆ THỐNG NHÚNG

1. HỆ THỐNG NHÚNG
2. PHẦN CỨNG HỆ THỐNG NHÚNG
- 3. PHẦN MỀM HỆ THỐNG NHÚNG**
4. CÔNG NGHỆ Multi-Core, SoC, and FPGA
5. MỘT SỐ ỨNG DỤNG HỆ THỐNG NHÚNG